

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
3	(a)			Indicative content: <u>Experimental set-up</u> Source clear in diagram or description Detector clear in diagram or description Distance clear in diagram or description. Blocking alpha and beta [with aluminium]. <u>Data collection</u> Vary distance. Measure counts for set time / long time / repeats Check for background With no source present <u>Analysis</u> Adjust for the background intensity (or nothing if they have said that the source is a strong source / lead container). Correct plot chosen e.g. intensity vs r^{-2}, or $(\text{count})^{-1/2}$ vs r $\ln$ (intensity or count) vs $\ln r$ Correct expected shape i.e. straight line Added detail e.g. error bars, intercept etc	6			6		6